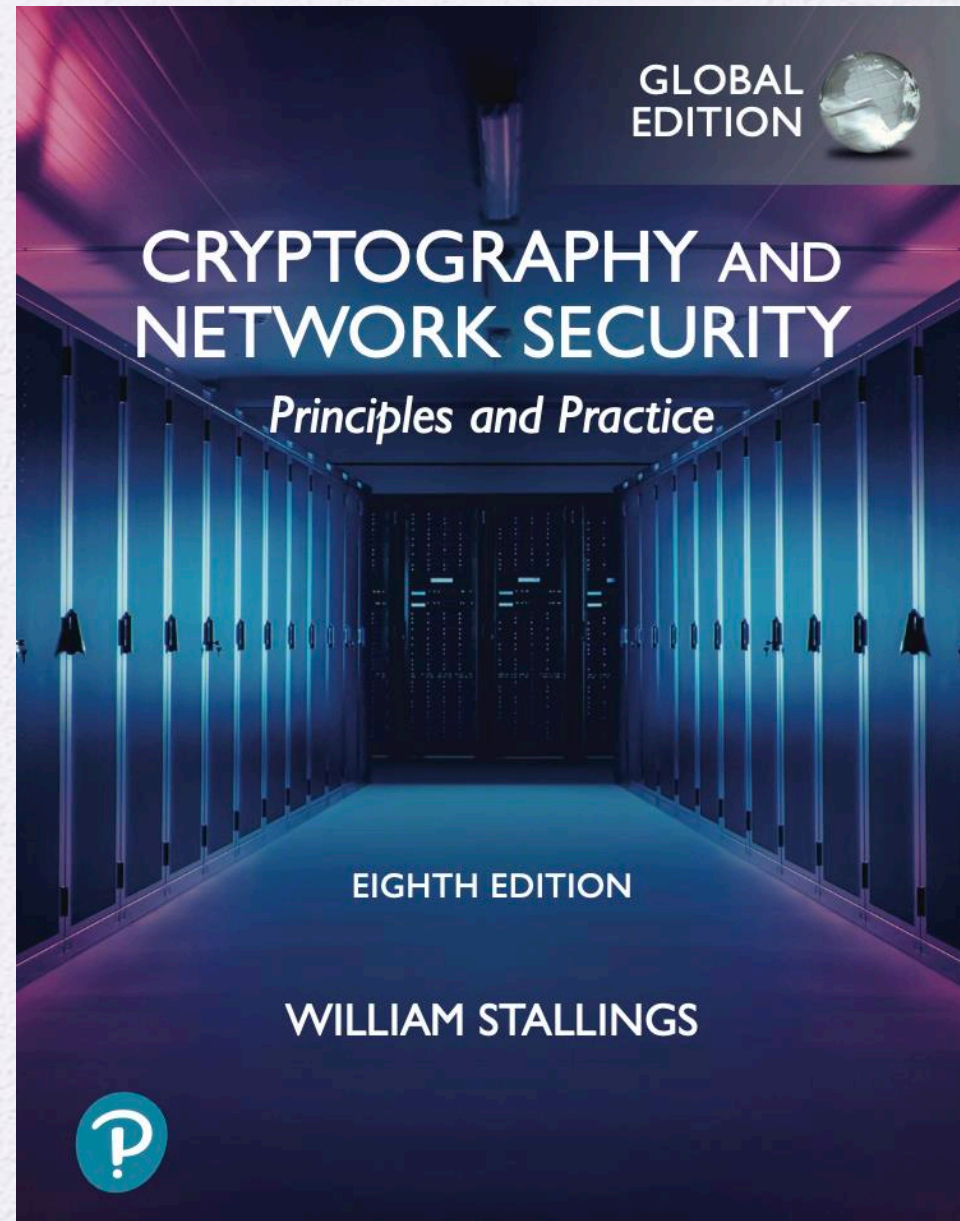


University of Nevada – Reno
Computer Science & Engineering
Department

CS454/654 Reliability and Security
of Computing Systems - Fall 2024



CLASSICAL ENCRYPTION TECHNIQUES

3.1 Symmetric Cipher Model

- Cryptography
- Cryptanalysis and Brute-Force Attack

3.2 Substitution Techniques

- Caesar Cipher
- Monoalphabetic Ciphers
- Playfair Cipher
- Hill Cipher
- Polyalphabetic Ciphers
- One-Time Pad

3.3 Transposition Techniques

3.4 Key Terms, Review Questions, and Problems

Definitions

Plaintext

- An original message

Ciphertext

- The coded message

Encryption

- The process of converting from plaintext to ciphertext

Decryption

- Restoring the plaintext from the ciphertext

Cryptography

- The area of study of the many schemes used for encryption

Cryptographic system/cipher

- A scheme (a specific method)

Cryptanalysis

- Techniques used for decrypting a message without any knowledge of the enciphering details

Breaking the
code →

Cryptology

- The areas of cryptography and cryptanalysis

Symmetric Cipher Model

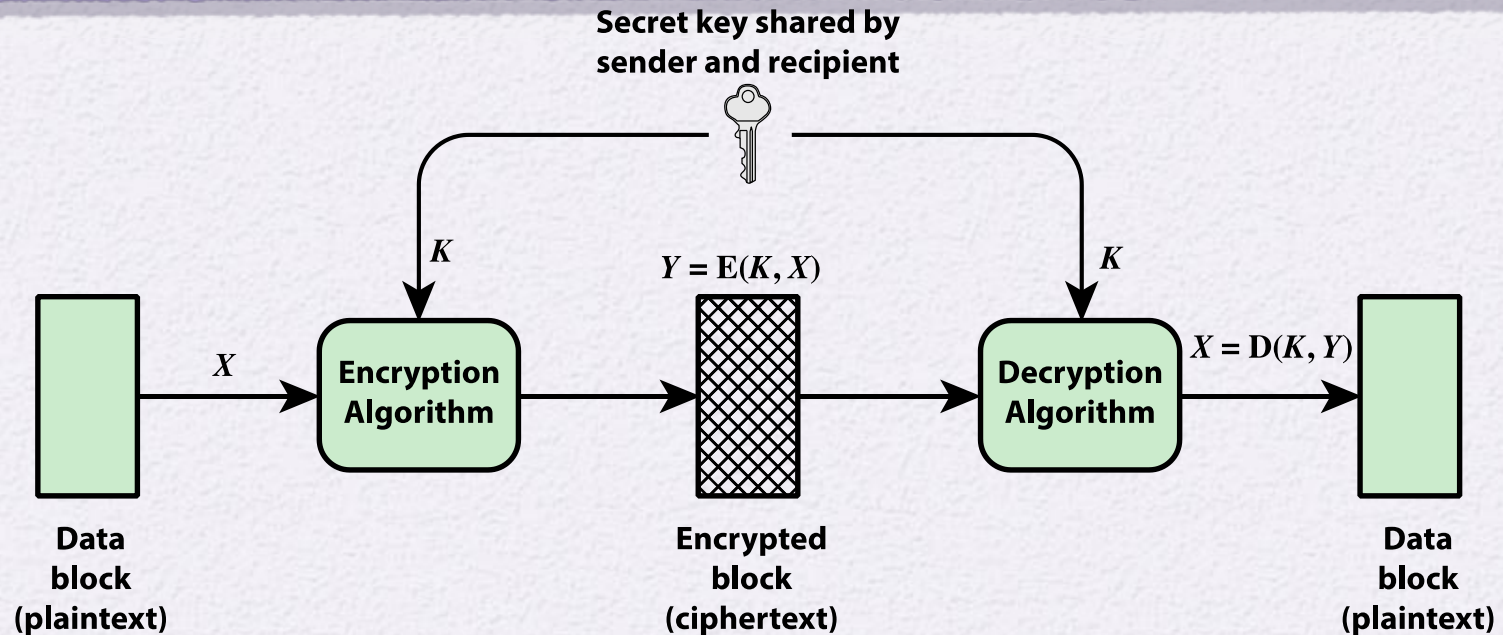
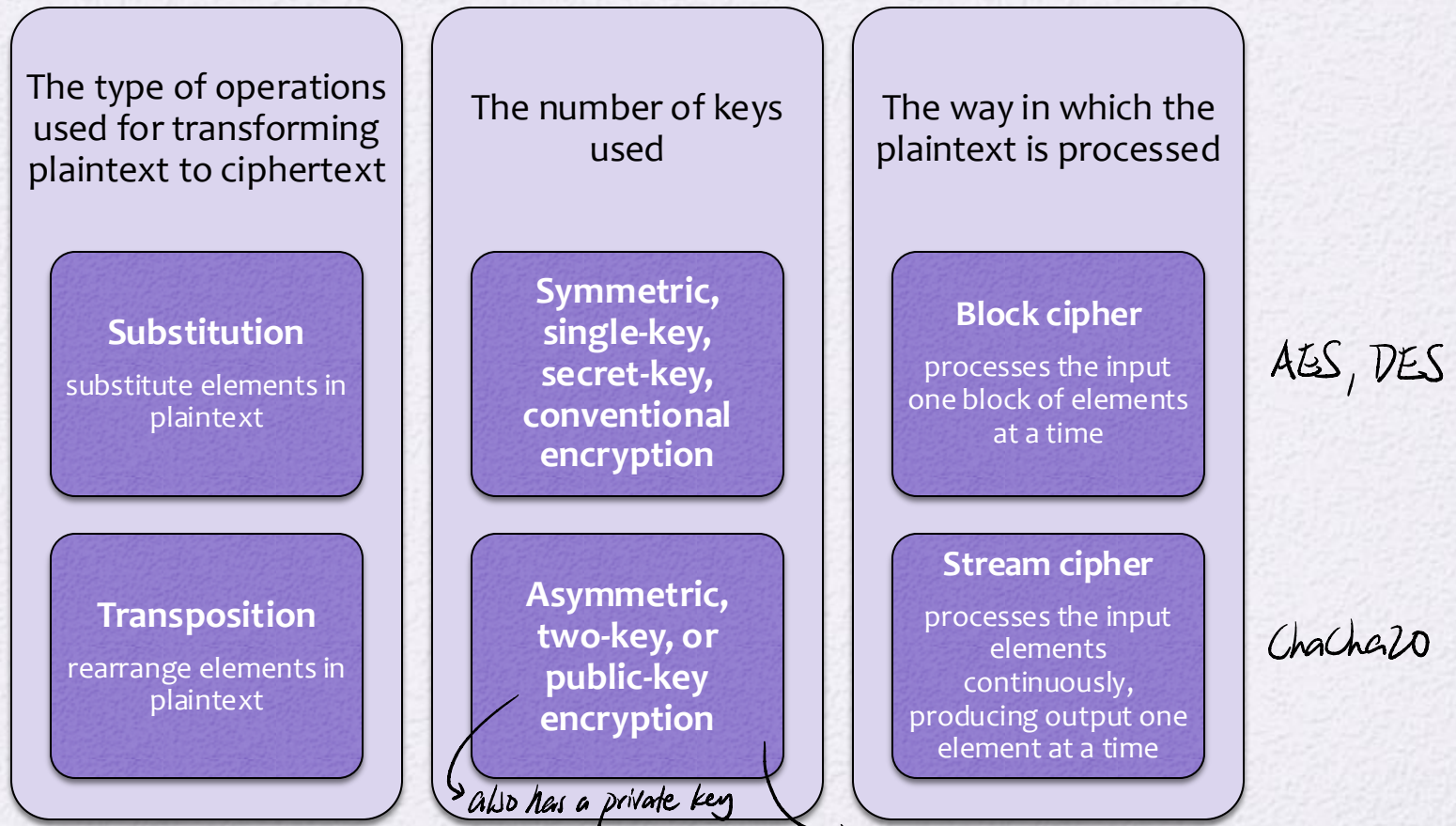


Figure 3.1 Simplified Model of Symmetric Encryption

- There are two requirements for secure use of conventional encryption:
 - A strong encryption algorithm
 - Sender and receiver must have **obtained copies of the secret key** in a secure fashion and must keep the key secure

Cryptographic Systems

- Characterized along three independent dimensions: *covered in chapter 1*



Most product systems involve multiple stages of substitutions and transpositions. *SSH keys*

Cryptanalysis and Brute-Force Attack

Cryptanalysis

- Attack relies on **the nature of the algorithm** plus some knowledge of the **general characteristics of the plaintext**
- Attack exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or to deduce the key being used

Brute-force attack

- Attacker tries **every possible key** on a piece of ciphertext until an intelligible translation into plaintext is obtained
- On average, **half of all possible** keys must be tried to achieve success

To supplement the brute-force approach, some **degree of knowledge about the expected plaintext is needed**, and some means of **automatically** distinguishing plaintext from garble is also needed

Table 3.1
Types of
Attacks
on
Encrypted
Messages

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • The analyst may be able to capture one or more plaintext messages as well as their encryptions.
Chosen Plaintext	• Encryption algorithm • Ciphertext • If the analyst is able somehow to get the source system to insert into the system a message chosen by the analyst.
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

(Table is on page 68 in the textbook)

Encryption Scheme Security

- **Unconditionally secure**

Definition

- No matter how much time and ciphertext an opponent has, it is impossible for him or her to decrypt the ciphertext.

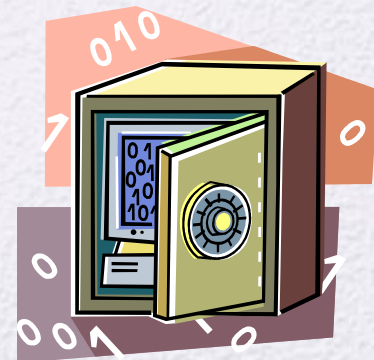
Reality

- With the exception of a scheme known as the one-time pad (described later in this chapter), there is **no encryption algorithm that is unconditionally secure.**

- **Computationally secure** *decryption takes computing time, with or w/o the key*

exp. time-consuming

- The cost of breaking the cipher exceeds the value of the encrypted information
- The time required to break the cipher exceeds the useful lifetime of the information



Strong Encryption

- The term **strong encryption** refers to encryption schemes that make it impractically difficult for unauthorized persons or systems to gain access to plaintext that has been encrypted
- Properties that make an encryption algorithm strong are:
 - Appropriate **choice of** cryptographic **algorithm**
 - Use of sufficiently **long key** lengths
 - Appropriate **choice of protocols**
 - A well-engineered implementation
 - Absence of **deliberately introduced** hidden **flaws**

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Substitution Technique

- The letters (bits) of plaintext are **replaced** by other letters (bits) or by numbers or symbols

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Caesar Cipher



- Simplest and earliest known use of a substitution cipher
- Involves **replacing** each letter of the alphabet with the letter standing **three places further** down the alphabet
- Alphabet is wrapped around so that the letter following Z is A

plain: meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

KEY	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
1	oggv	og	chvgt	vjg	vqic	rctva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrpc	rfe	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	objv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlg
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fghjo
14	btti	bt	puitg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgrc	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnn	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzxx	znk	zumg	vgxze
24	rjyy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxv

Figure 3.3 Brute-Force Cryptanalysis of Caesar Cipher

Monoalphabetic Cipher

- Compared to Caesar Cipher, Monoalphabetic Cipher allows an **arbitrary substitution**.
- The cipher line can be **any permutation** of the 26 alphabetic characters
 - **Q: How many possible keys?**
 - 26! possible keys.
- Possible attack can be **analyzing the relative frequency of letters**, and compare it to frequency distribution for English.
 - Attacker should know the nature of plaintext (text is in English)
 - Should have long message to generate correct frequency distribution.

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

- **P** and **Z** are the equivalents of plain letters **e** and **t**
- The letters **S, U, O, M,** and **H** are all of relatively high frequency and probably correspond to plain letters from the set **{a, h, i, n, o, r, s}**

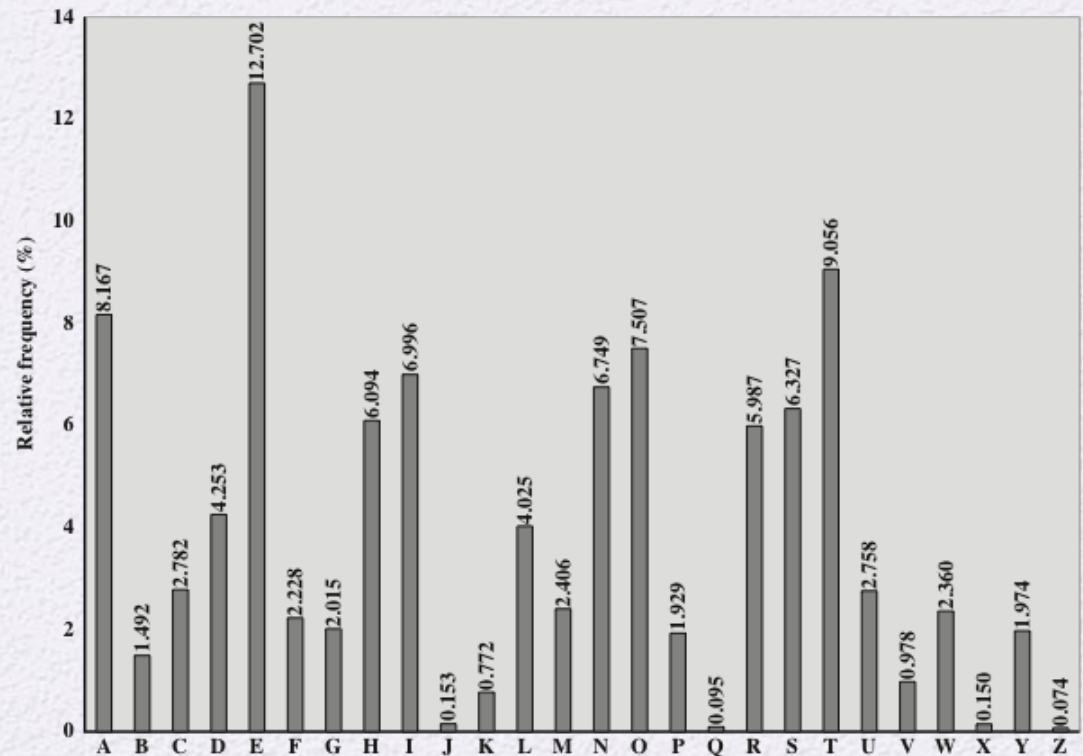
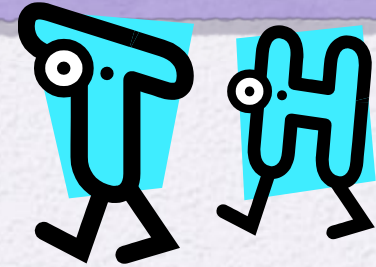


Figure 3.5 Relative Frequency of Letters in English Text

Monoalphabetic Ciphers

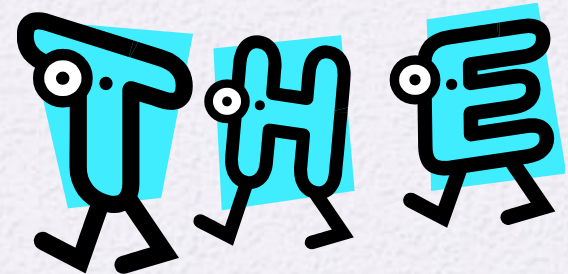
- Digram

- Two-letter combination
- Most common is *th*
- In our ciphertext, the most common digram is *ZW*, which appears three times. So we make the correspondence of *Z* with *t* and *W* with *h*.



- Trigram

- Three-letter combination
- Most frequent is *the*
- *ZWP* is most frequent trigram thus we can assume *P* correspond to *e*



- Monoalphabetic ciphers are *easy to break* because they *reflect the frequency data* of the original alphabet.

Playfair Cipher

- Best-known multiple-letter encryption cipher
- Treats **digrams** in the plaintext as **single units** and translates these units into ciphertext digrams
- Invented by British scientist Sir Charles Wheatstone in 1854
- Used as the standard field system by the British Army in World War I and the U.S. Army and other Allied forces during World War II

Playfair Key Matrix

Constructing 5x5 matrix from keyword.

1. Assume keyword is “jurisdiction”
2. Remove duplicates, keyword become “J U R I S D C T O N”
3. Treat I and J same, “J/I U R S D C T O N”
4. Fill the letters in keyword from left to right and then fill other letters in alphabetic order

J/I	U	R	S	D
C	T	O	N	A
B	E	F	G	H
K	L	M	P	Q
V	W	X	Y	Z

Playfair Key Matrix

Plaintext is encrypted **two letters at a time**, according to the following rules:

1. Repeating plaintext letters that are in the same pair are separated with a filler letter, such as **x**, so that **balloon** would be treated as **ba lx lo on**.
2. Two plaintext letters that **fall in the same row** of the matrix are each replaced by the letter to the **right**, with the first element of the row circularly following the last. For example, **ar** is encrypted as **RM**.
3. Two plaintext letters that **fall in the same column** are each replaced by the letter **beneath**, with the top element of the column circularly following the last. For example, **mu** is encrypted as **CM**.
4. Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, **hs** becomes **BP** and **ea** becomes **IM** (or JM, as the encipherer wishes). (**Create rectangle**).

Play → pl ay => QP NB

Game → ga me => IN CL

hell → he lx lx => CF SU SU

hello → he lx lo => CF SU PM

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Substitution Technique

- The letters (bits) of plaintext are **replaced** by other letters (bits) or by numbers or symbols

Substitution Techniques

- Caesar Cipher
- Monoalphabetic Ciphers
- Playfair Cipher
- **Hill Cipher**
- Polyalphabetic Ciphers
- One-Time Pad



Hill Cipher

- Take $n \times n$ (3×3 , 2×2 , etc.) matrix which should be invertible.
- Inverse M^{-1} of square matrix M is defined by the equation $M(M^{-1}) = (M^{-1})M = I$ where I is identity matrix.

$$\mathbf{C} = E(\mathbf{K}, \mathbf{P}) = \mathbf{PK} \bmod 26$$

$$\mathbf{P} = D(\mathbf{K}, \mathbf{C}) = \mathbf{CK}^{-1} \bmod 26 = \mathbf{PKK}^{-1} = \mathbf{P}$$

Hill Cipher

$$\mathbf{K} = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \quad \mathbf{K}^{-1} = \begin{pmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{pmatrix} \quad \begin{aligned} \mathbf{C} &= \mathbf{E}(\mathbf{K}, \mathbf{P}) = \mathbf{PK} \bmod 26 \\ \mathbf{P} &= \mathbf{D}(\mathbf{K}, \mathbf{C}) = \mathbf{CK}^{-1} \bmod 26 = \mathbf{PKK}^{-1} = \mathbf{P} \end{aligned}$$

Example: consider plaintext “paymoremoney” and use the encryption key above.

First three letters (pay) of the plaintext are represented by the vector (15 0 24)

Then $(15 \ 0 \ 24) \mathbf{K} = (303 \ 303 \ 531) \bmod 26 = (17 \ 17 \ 11) = \text{RRL}$. And if we continue “paymoremoney” $\rightarrow \text{RRLMWBKASPDH}$

For decryption if we repeat the process for RRLMWBKASPDH with $\mathbf{K}^{-1}$ we get the “paymoremoney”.

Substitution Technique

- The letters (bits) of plaintext are **replaced** by other letters (bits) or by numbers or symbols

Substitution Techniques

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Polyalphabetic Ciphers

- Polyalphabetic substitution cipher
 - Improves on the simple monoalphabetic technique by using different monoalphabetic substitutions as one proceeds through the plaintext message
- Polyalphabetic Ciphers: Vigenere and Vernam

Vigenère Cipher

- To encrypt a message, a key is needed that is **as long as the message**
- Usually, the key is a repeating keyword
- For example, if the keyword is **deceptive**, the message “**we are discovered save yourself**” is encrypted as:

key: deceptivedeceptivedeceptive

plaintext: wearediscoveredsaveyourself

ciphertext: Z|C|V|T|W|Q|N|G|R|Z|G|V|T|W|A|V|Z|H|C|Q|Y|G|L|M|G|J

$$d+w=z \Rightarrow (3+22) \bmod 26=25$$

with $a=0$, $z=25$.

Vigenère Cipher

key	3	4	2	4	15	19	8	21	4	3	4	2	4	15
plaintext	22	4	0	17	4	3	8	18	2	14	21	4	17	4
ciphertext	25	8	2	21	19	22	16	13	6	17	25	6	21	19

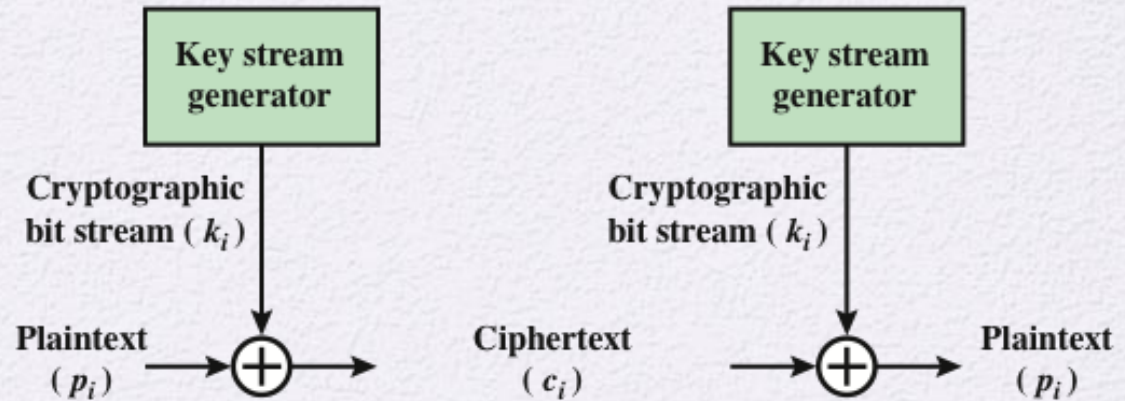
In the example, two instances of the sequence “red” are separated by **nine** character positions.

If the **message is long enough**, there **will be a number of such repeated ciphertext sequences**.

By looking for common factors in the displacements of the various sequences, the analyst should be able to make a good guess of the keyword length.

Vernam Cipher

Key can be shorter or equal to message, and can be reused.



$$c_i = p_i \oplus k_i$$

Figure 3.7 Vernam Cipher

where

p_i = i th binary digit of plaintext

k_i = i th binary digit of key

c_i = i th binary digit of ciphertext

$\oplus$ = exclusive-or (XOR) operation

Key is just a random bits

One-Time Pad

- Improvement to Vernam cipher proposed by an Army Signal Corp officer, Joseph Mauborgne
- Use a **random key** that is as **long as the message** so that the key need **not be repeated**
- Key is used to encrypt and decrypt a **single message** and then is **discarded**
- Each **new message** requires a **new key** of the same length as the new message
- Produces random output that bears no statistical relationship to the plaintext



Difficulties

- The one-time pad **offers complete security** but, in practice, has two fundamental difficulties:
 - There is the practical **problem** of making **large quantities of random keys**
 - Any heavily used system might require millions of random characters on a regular basis
 - Mammoth key distribution problem
 - For every message to be sent, a key of equal length is needed by both sender and receiver. Thus, distribution (exchanging) of keys is challenging.

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Rail Fence Cipher

- Simplest **transposition** cipher
- Plaintext is **written** down as a **sequence of diagonals** and then **read** off as a **sequence of rows**
- To encipher the message “meet me after the toga party” with a rail fence of **depth 2**, we would write:

m e m a t r h t g p r y m e t m e t h t o s p a r y **depth 3**
 ↓ ↓ ↓ ↓ ↓ e a e a t
 e t e f e t e o a a t

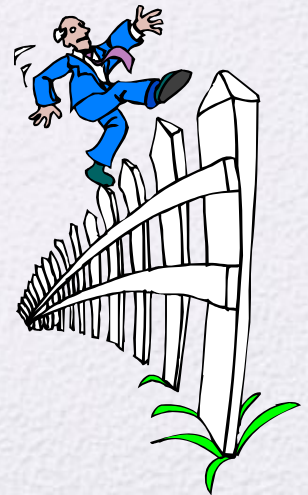
m m t o a e l e t h t g p r y e a e a t

Encrypted message is: → Z m t o a e l e b t g p r y Z (e a) t

MEMATRHTGPRYETEFETEOAAT

Q: What is key here?

The Depth: number of rows



Row Transposition Cipher

- Write the message in a rectangle, row by row, and read the message off, column by column, **but permute the order of the columns**
 - The **order of the columns then becomes the key** to the algorithm

Key:	4	3	1	2	5	6	7
Plaintext:	a	t	t	a	c	k	p
	o	s	t	p	o	n	e
	d	u	n	t	i	l	t
	w	o	a	m	x	y	z
Ciphertext:	TTNAAPTMTSUOAODWCOIXKNLYPETZ						

The transposition cipher can be made significantly more secure by performing **more than one stage of transposition**